Aftab Hossain Mikat

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About

Senior Computer Science and Engineering undergraduate at Rajshahi University of Engineering and Technology (RUET). I am exploring multimodal AI, vision-language models, and NLP, especially for Bangla and other low-resource languages. I also study compact language models, data-efficient learning, and training methods that help smaller models learn effectively.

Education

Rajshahi University of Engineering and Technology (RUET) *Dec 2022 – Feb 2027*
B.Sc. in Computer Science and Engineering (Expected) *Rajshahi, Bangladesh*

- **CGPA:** 3.43/4.00 through the 7th semester. Average across the previous two semesters: 3.68/4.00.
- **Focus:** Natural Language Processing, Small Language Models, Deep Learning, Algorithms, and Software Engineering.

St. Joseph Higher Secondary School *Jul 2019 – Jun 2022*
Higher Secondary *Dhaka, Bangladesh*

- **GPA:** 5.00/5.00.

Publications

Dense Supervision, Gentle Decay: Curriculum-Guided Rebalancing of ELECTRA-Style RTD and MLM for Small-Data Pretraining *Aug 2026*
Aftab Hossain Mikat, Tamjid Hasan Fahim
Accepted workshop paper at BabyLM Workshop @ EMNLP 2026 – Strict-Small Track *Preprint*
Studies curriculum-guided RTD–MLM rebalancing and training strategies for compact Bengali encoders under a strict data budget.

When the Evidence Is Right, the Math Is Easy: Understanding the Challenges of Document-Grounded Reasoning *Sep 2026*
Aftab Hossain Mikat, Tamjid Hasan Fahim
Accepted archival workshop paper (poster) at DocInsights Workshop @ EMNLP 2026
DocSem Shared Task Preprint
An analysis of document-grounded numerical reasoning, showing how visual evidence extraction and localization can be harder than the arithmetic itself.

Experience

Walton Hi-Tech Industries PLC *Feb 2026 – Mar 2026*
ICT Department - Two-week Industrial Attachment *Gazipur, Bangladesh*

- Gained practical exposure to enterprise software, server infrastructure, networking, manufacturing technology, and quality-control operations across industrial departments.
- Collaborated in a team to build a Notification Management Website using React, TypeScript, Vite, and Django; the system received positive feedback from the organization.

Projects

BoostLearning AI: AI-Powered Study Assistant *GitHub*

- Developed a full-stack study assistant featuring a customizable lab-report generator, PDF/text-to-MCQ and narrative question generation, topic explanation, paraphrasing, and personalized study planning.
- **Tech Stack:** React, TypeScript, Flask, Firebase Authentication, MongoDB, Gemini API

Notification Management Website *GitHub*

- Built with a team during the Walton industrial attachment to support structured creation and management of organizational notifications through a modern web interface.
- **Tech Stack:** React, TypeScript, Vite, Django

WhatsApp Chat Analyzer: Communication Analytics Platform *GitHub*

- Developed a web-based communication analytics platform with NLP/ML-powered sentiment analysis, message classification, multilingual support, interactive visualizations, and secure JWT/Bcrypt authentication; optimized large-chat processing through caching and batch translation.
- **Tech Stack:** Python, Streamlit, NLP/ML, JWT, Bcrypt

MovieZone: Movie Discovery Platform *Live – GitHub*

- Developed a React and TypeScript movie discovery platform with search, genre and country filtering, ratings-based sorting, detailed movie information, personalized watchlists, recommendations, and dark mode. Integrated an external API for rich film and series data.
- **Tech Stack:** React, TypeScript, External Movie API

Full-Stack Blog Platform *Live – GitHub*

- Developed a full-stack blogging platform for publishing and reading long-form content with a responsive user experience and persistent data storage.
- **Tech Stack:** Node.js, React, TypeScript, MongoDB

Technical Skills

Programming	C++, Python, JavaScript, TypeScript
AI and ML	PyTorch, Hugging Face Transformers, Scikit-learn, Pandas, NumPy, NLP, Computer Vision, Multimodal AI, Small Language Models, Deep Learning
Web	React, Vite, Node.js, Express.js, Flask, Django, HTML, CSS, Bootstrap
Databases	MongoDB, MySQL, Firebase
Tools	Git, GitHub, LaTeX, Kaggle, Vercel, Railway
CS Foundations	Data Structures and Algorithms, Competitive Programming, Algorithmic Problem Solving

Research Interests and Achievements

Research Interests: Multimodal AI, Vision-Language Models, Efficient NLP, Small Language Models, Knowledge Distillation, Teacher–Student Learning, Bangla and Low-resource Language AI, and Document-Grounded Reasoning.

Achievements: Two accepted EMNLP 2026 workshop papers: BabyLM (Strict-Small Track) and DocInsights (DocSem Shared Task, archival poster).

Additional Achievements

- Solved 250+ programming problems on Codeforces.
- Received General Scholarships in the Secondary School Certificate and Junior School Certificate examinations.
- **Languages:** Bengali and English.